TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

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Warning

- **A hot engine and intake air system can cause severe burns. Turn off the engine and wait until they are cool before removing the intake-air system.**

Caution

- **Perform exhaust gas temperature sensor removal/installation only when inspection or replacement is required, such as when DTCs for the sensor are displayed, because it may cause exhaust gas leakage. If the exhaust gas temperature sensor is removed once, always verify that there is no exhaust gas leakage after reinstallation.**

Note

- When removing or replacing the variable geometry turbocharger, perform the “Operation After Replacing or Removal/Installation Turbocharger”.

Operation After Replacing or Removal/Installation Turbocharger

1. If the turbocharger is replaced or removed/installed, perform the following procedure.

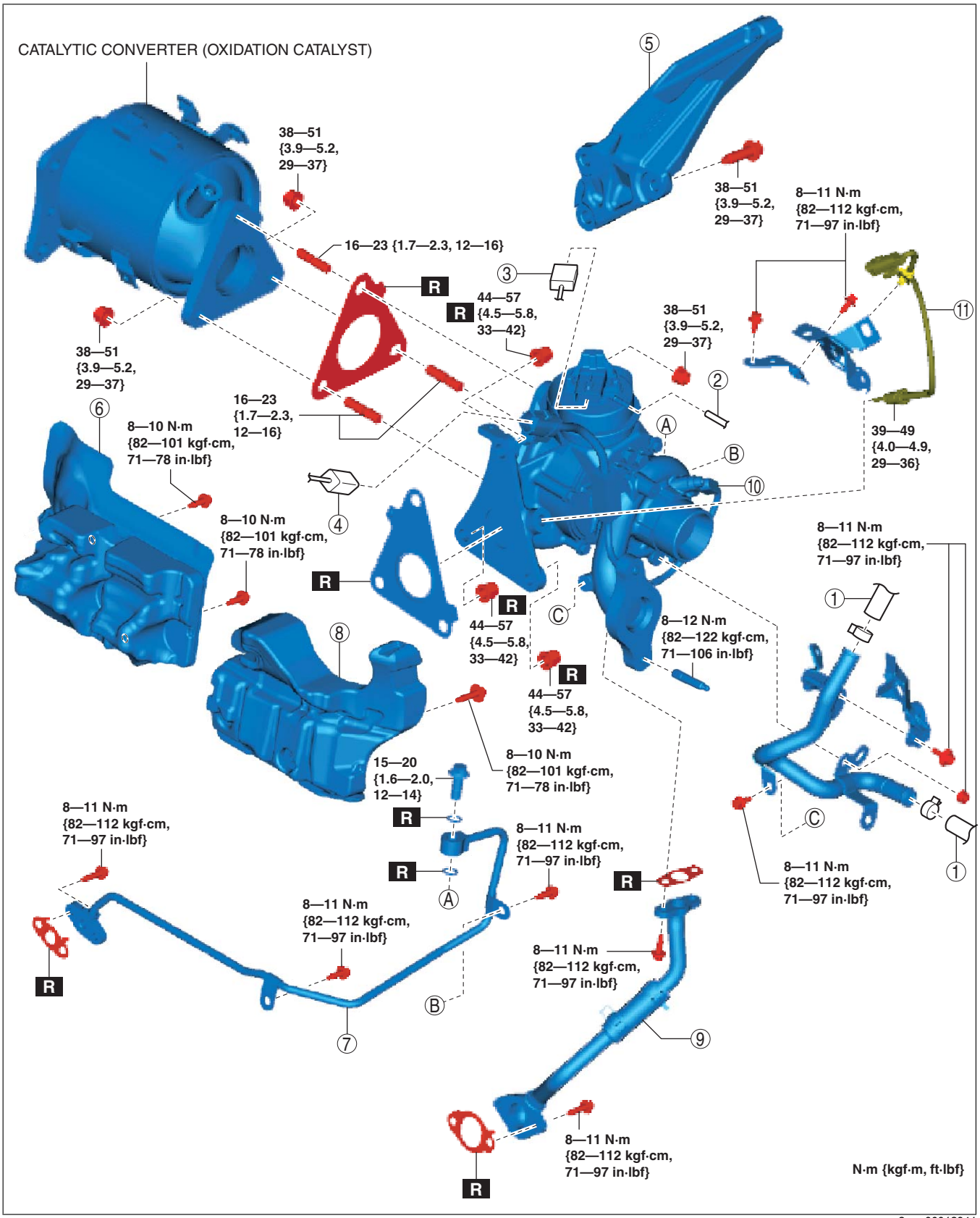
STEP	ACTION	PAGE/CONDITION
1	Perform turbocharger initialization procedure.	(See TURBOCHARGER INITIALIZATION [SKYACTIV-D 1.5].)
2	Switch the ignition off.	—
3	Wait for 20 s.	—
4	Start the engine.	—
5	Maintain the engine idling condition for 10 s.	—
6	Switch the ignition off.	—
7	Wait for 20 s.	—
8	Perform KOEO self-test procedure.	(See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].)
9	Perform KOER self-test procedure.	(See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].)

Turbocharger With Variable Turbine Geometry Removal/Installation

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
2. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Drain the engine coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
4. Remove the following parts: (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   - Fresh-air duct
   - Air cleaner
   - Air hose
   - Air inlet pipe
5. Remove the battery and the battery tray. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
6. Remove the following parts as a single unit: (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   - Turbocharger air inlet hose
   - Blow-by heater
   - Breather hose
7. Remove the turbocharger air outlet pipe. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
8. Remove the front crossmember. (See FRONT CROSSMEMBER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
9. Remove the catalytic converter (DPF). (See EXHAUST SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
10. Remove in the order shown in the figure.
11. Install in the reverse order of removal.
12. Refill the engine coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
13. Perform the following procedure.

    When newly replacing Turbocharger With Variable Turbine Geometry
    - Replace the engine oil. (See ENGINE OIL REPLACEMENT [SKYACTIV-D 1.5].)

    When installing a removed Turbocharger With Variable Turbine Geometry
    - Inspect the engine oil level. (See ENGINE OIL LEVEL INSPECTION [SKYACTIV-D 1.5].)



1	Heater hose
2	Vacuum hose (See Vacuum hose installation note.)
3	Turbocharger actuator connector
4	Speed sensor assy connector
5	Turbocharger bracket
6	Insulator

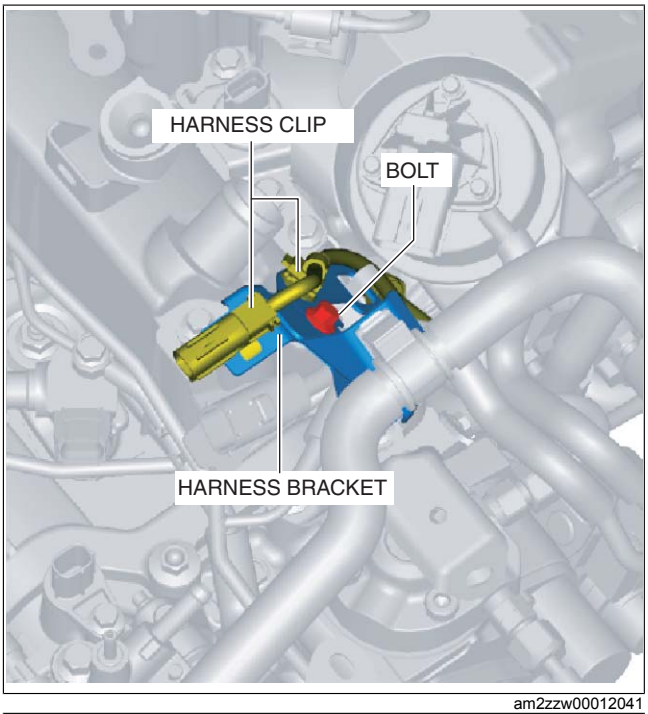
7	Oil main pipe (See Oil main pipe removal note.) (See Oil main pipe installation note.)
8	Insulator
9	Oil return pipe

10	Turbocharger with variable turbine geometry (See Operation After Replacing or Removal/ Installation Turbocharger.)

11	Exhaust gas temperature sensor No.1

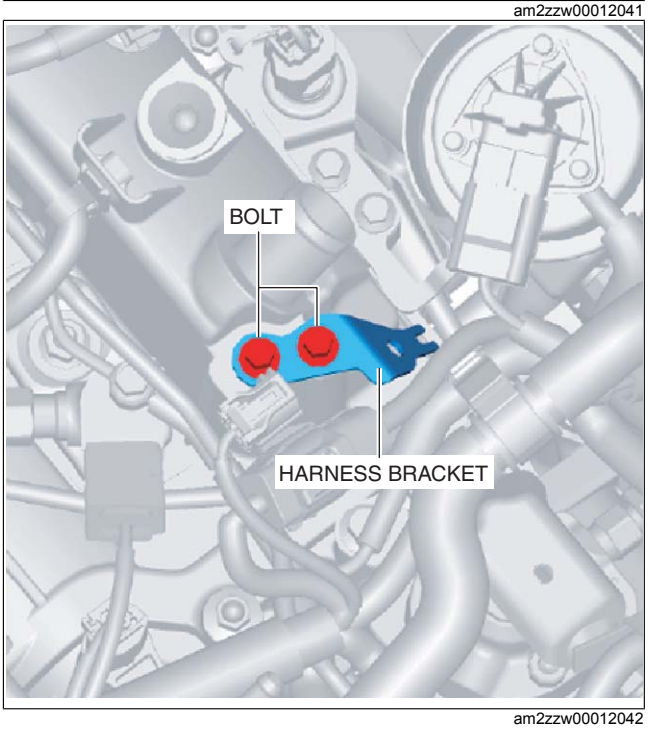
Oil main pipe removal note

1. Disconnect the exhaust gas temperature sensor No.1. (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
2. Remove the wiring harness bracket shown in the figure.
3. Disconnect the exhaust gas pressure sensor No.1. (See EXHAUST GAS PRESSURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



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4. Remove the wiring harness bracket shown in the figure.
5. Remove the oil main pipe

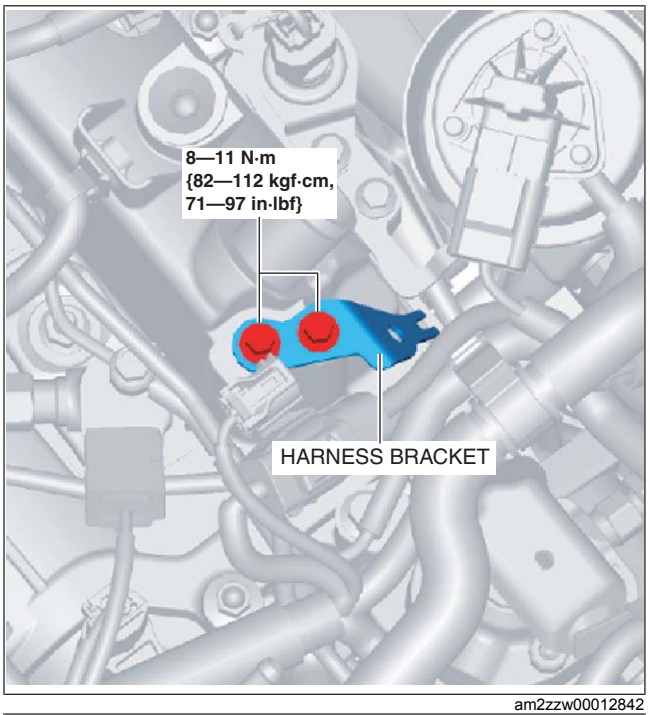


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Oil main pipe installation note

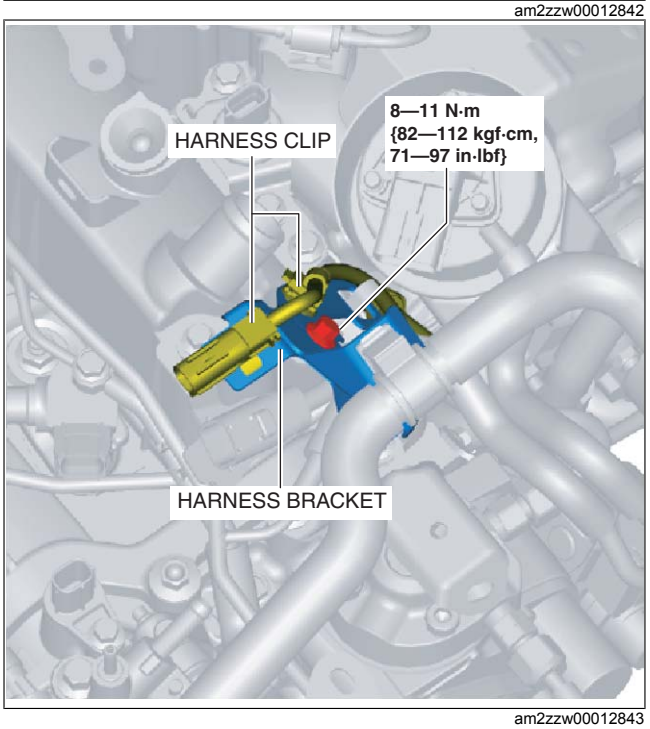
1. Install the oil main pipe.

2. Install the harness bracket as shown in the figure.
3. Connect the exhaust gas pressure sensor No.1 connector. (See EXHAUST GAS PRESSURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



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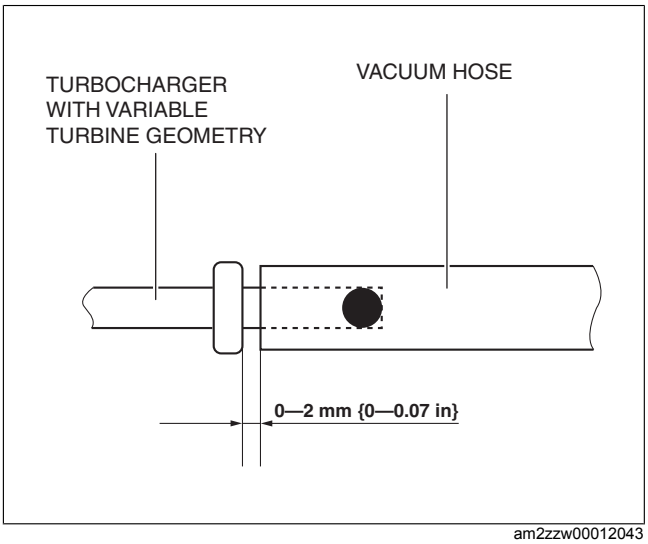
4. Install the harness bracket as shown in the figure.
5. Connect the exhaust gas temperature sensor No.1 connector. (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



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Vacuum hose installation note

1. Install the vacuum hose as shown in the figure.



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